

The empirical recurrence for OEIS A235102: recovering a conjecture that is not written down

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Abstract

OEIS A235102 records “Empirical recurrence of order 62”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 1512$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 62, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 62 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235102 is “Number of $(n+1) \times (3+1)$ 0..6 arrays with every 2×2 subblock having its diagonal sum differing from its antidiagonal sum by 4, with no adjacent elements equal (constant-stress tilted 1×1 tilings)”. It has offset 1 and begins

2900, 9328, 30312, 101464, 345280, 1201872, 4256548, 15381300, ...

The entry states:

Empirical recurrence of order 62 (see link above).

As of the “Last modified” line on the live entry (Jun 19 2022 21:18:43, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1512$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 1512$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 3024$ exact terms of the model returns order 62 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{62} c_i a(n-i),$$

c_1	26	c_{17}	99477066094	c_{33}	178350675584016536	c_{49}	127771419687920160
c_2	−197	c_{18}	5830990002604	c_{34}	−72870132047722387	c_{50}	−92944441829847712
c_3	−618	c_{19}	−5531103911050	c_{35}	−334019463522327456	c_{51}	−45529214722506880
c_4	16236	c_{20}	−36333941474477	c_{36}	181312841289301481	c_{52}	30759359807986560
c_5	−47968	c_{21}	59670835278754	c_{37}	521378612885307758	c_{53}	12047539637251328
c_6	−450077	c_{22}	173933326235209	c_{38}	−335472819876389500	c_{54}	−7411485795664384
c_7	3040508	c_{23}	−414795761909096	c_{39}	−677646064725427976	c_{55}	−2246498452366336
c_8	3834105	c_{24}	−636203937377429	c_{40}	484884754203078548	c_{56}	1241737010706432
c_9	−80993358	c_{25}	2153597061702698	c_{41}	730798943578300580	c_{57}	272580360351744
c_{10}	94169994	c_{26}	1707879859184080	c_{42}	−556930197747557422	c_{58}	−134934618193920
c_{11}	1274573308	c_{27}	−8810365700187812	c_{43}	−649850383015293540	c_{59}	−18749490020352
c_{12}	−3626150019	c_{28}	−2872296323611493	c_{44}	510043760274672416	c_{60}	8394557423616
c_{13}	−12505213618	c_{29}	29159022031552970	c_{45}	471966380356851064	c_{61}	532338573312
c_{14}	62450102449	c_{30}	101580543883199	c_{46}	−370657370123719680	c_{62}	−220278030336
c_{15}	65511304866	c_{31}	−79240705755251326	c_{47}	−276177355808278160		
c_{16}	−707970519435	c_{32}	18363404418339988	c_{48}	211363609308415392		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 62, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $62 + 4$ terms removed returns order 62 as well, so $\deg q_{\min} = 62 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $62 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 62 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A235102.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.